



Ever Wondered What Keeps Your Food From Growing Mold?

Breads and pastries are easily spoiled by molds and yeasts. Because they contain moisture and nutrients, they can become a good environment for microorganisms to grow.

But there's a compound that helps keep them at bay.

POTASSIUM SORBATE

Potassium sorbate is a potassium salt having sorbate as the counterion. It has a role as an antimicrobial food preservative.

POTASSIUM SORBATE

dissolves ↓ in food
SORBIC ACID

enters fungal cells

interferes with enzymes & cell membranes

slows metabolism, growth & reproduction

It is a preservative action more efficient in acidic foods.

WHY? Potassium sorbate exists in equilibrium with sorbic acid. At lower pH, more sorbic acid is present in its uncharged form, allowing it to enter fungal cells and interfere with their growth.

Sorbic acid + K⁺ ⇌ Potassium sorbate

Potassium sorbate is the potassium salt of sorbic acid.

Molecular Formula:

C₆H₇O₂K

IUPAC Name:

potassium (2E,4E)-hexa-2,4-dienoate

Molar Mass:

150.22 g/mol

Molecular Formula:

C₆H₇O₂K

Physical Description:

Liquid; CBI; Dry Powder

Liquid; Other Solid; Dry Powder

White solid; [HSDB]

Color / Form:

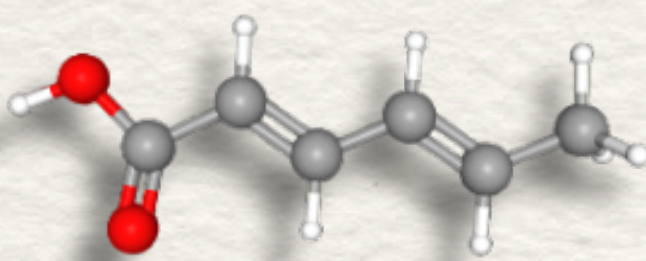
White powder

Crystals

White crystals or powder

Odor:

Characteristic odor



Melting Point:

for sorbic acid, Melting range of sorbic acid isolated by acidification and not recrystallised 133 °C to 135 °C after vacuum drying in a sulphuric acid desiccator >270 °C (decomposes)

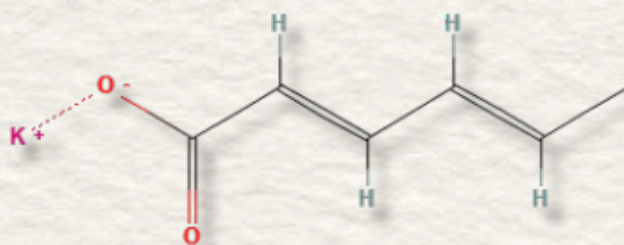
Solubility:

Solubility in water at 20 °C:

58.2%; in alcohol: 6.5%

Soluble in water at 25 °C

58.5 g in 100 cc water at 25 °C



MOLECULAR STRUCTURE:

CH₃-CH=CH-CH=CH-COO⁻ K⁺

CARBOXYLATE GROUP

The -COO⁻K⁺ group makes potassium sorbate the potassium salt of sorbic acid.

CONJUGATED DOUBLE BONDS

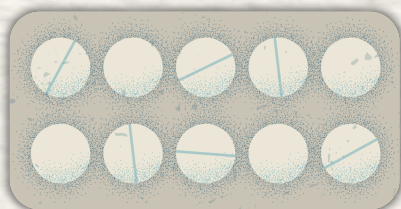
Its C=C-C=C system gives the molecule a conjugated structure, an important feature of sorbic acid.

Its Other Uses Aside From Food Preservation



Used as an antimicrobial preservative in some cosmetic and personal-care formulations to **help control the growth of microorganisms**.

Can serve as a preservative in some formulations, helping **protect products from microbial contamination and spoilage**.



BUT FIRST, SAFETY

Dust may irritate the eyes, skin, and respiratory tract.

Not highly flammable under normal conditions.

Stable when properly stored and handled.

Avoid releasing large quantities into the environment.

Gloves + eye protection + avoid inhaling dust.

RELATION TO CHEMICAL ENGINEERING

Chemical engineers help determine the right concentration and distribution of potassium sorbate in a food product.

The formulation must provide enough preservative to inhibit mold and yeast while maintaining the product's taste, texture, pH, and shelf life.

Potassium sorbate > Controlled formulation > Effective antimicrobial concentration > Longer-lasting product + maintained quality

REFERENCES:

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